Participant Details

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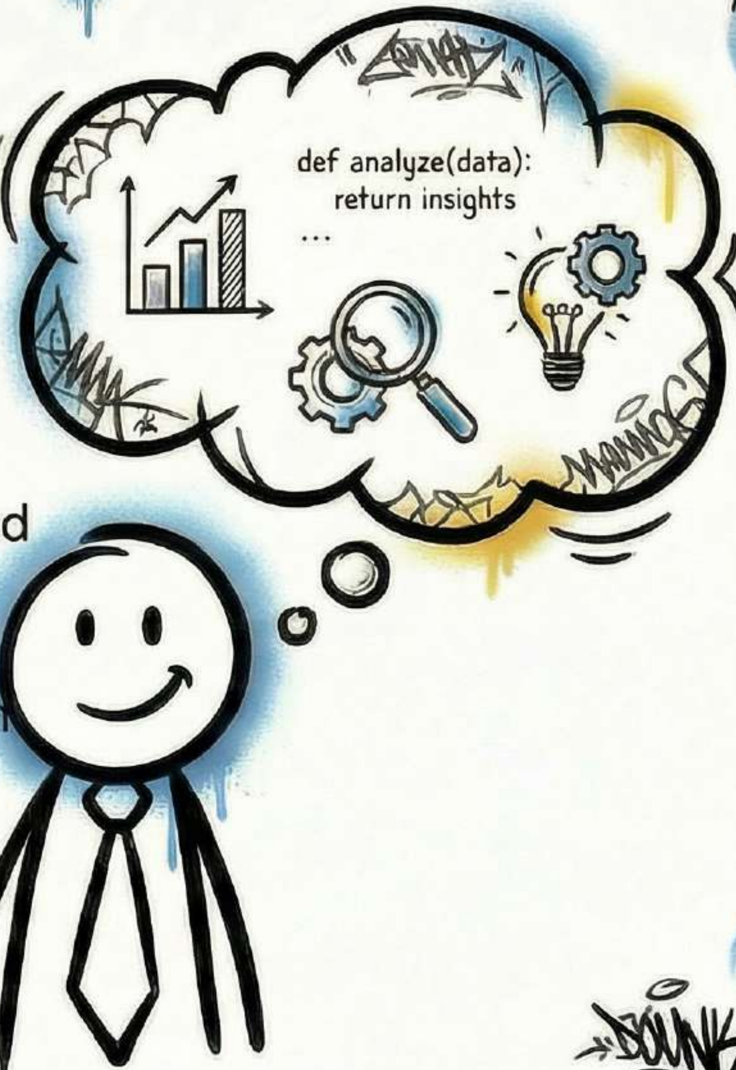
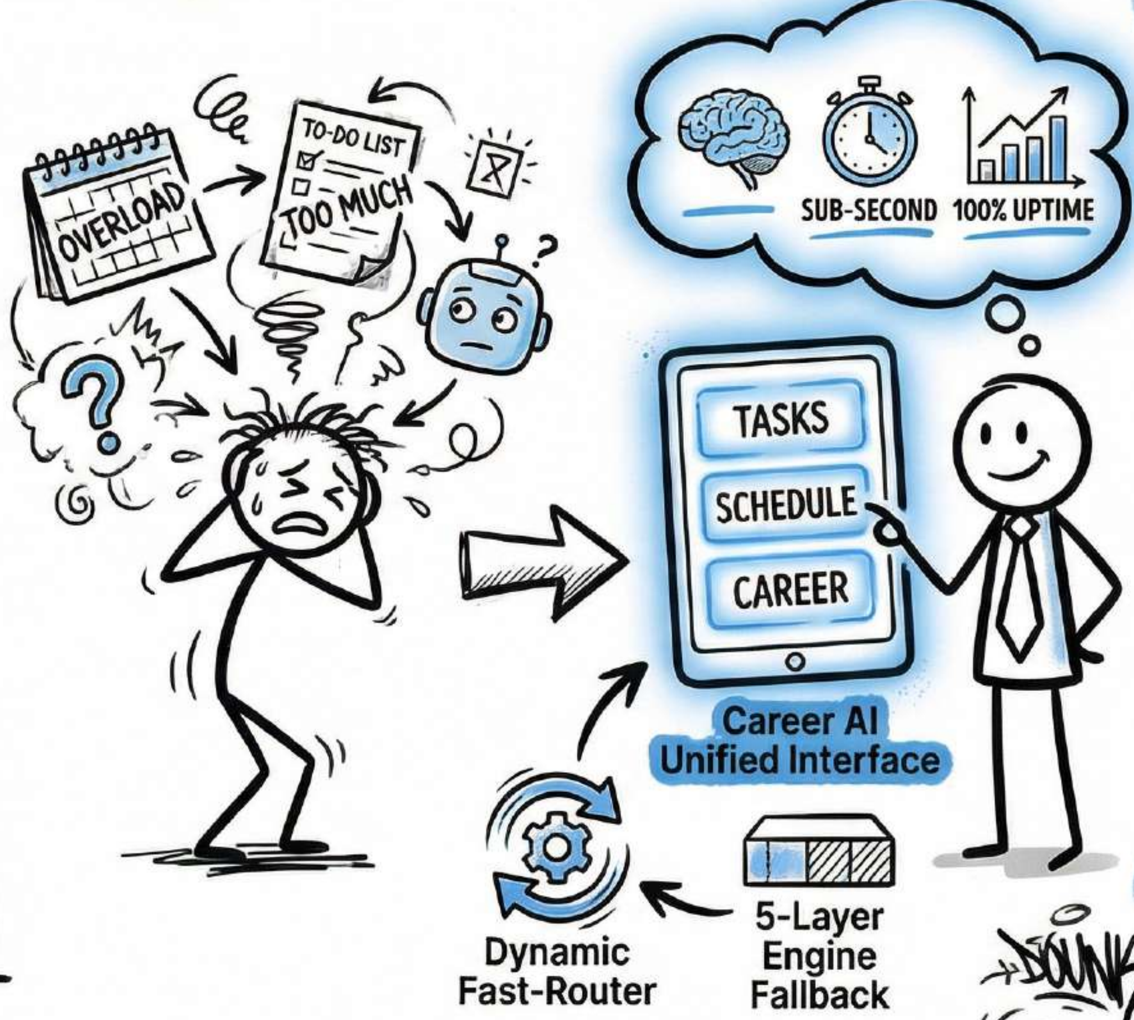
- **Background:** BBA student in Business Analytics at St. Aloysius College, Mangalore. Born in Abu Dhabi, UAE.
- **Expertise:** Data Visualization, Python, Critical Thinking, Communication, Organizational Planning, Project Management, Adaptability.
- **Portfolio:** https://jts.dpdns.org/

Problem Statement:
Career AI: Unified Multi-Agent Personal & Career Platform



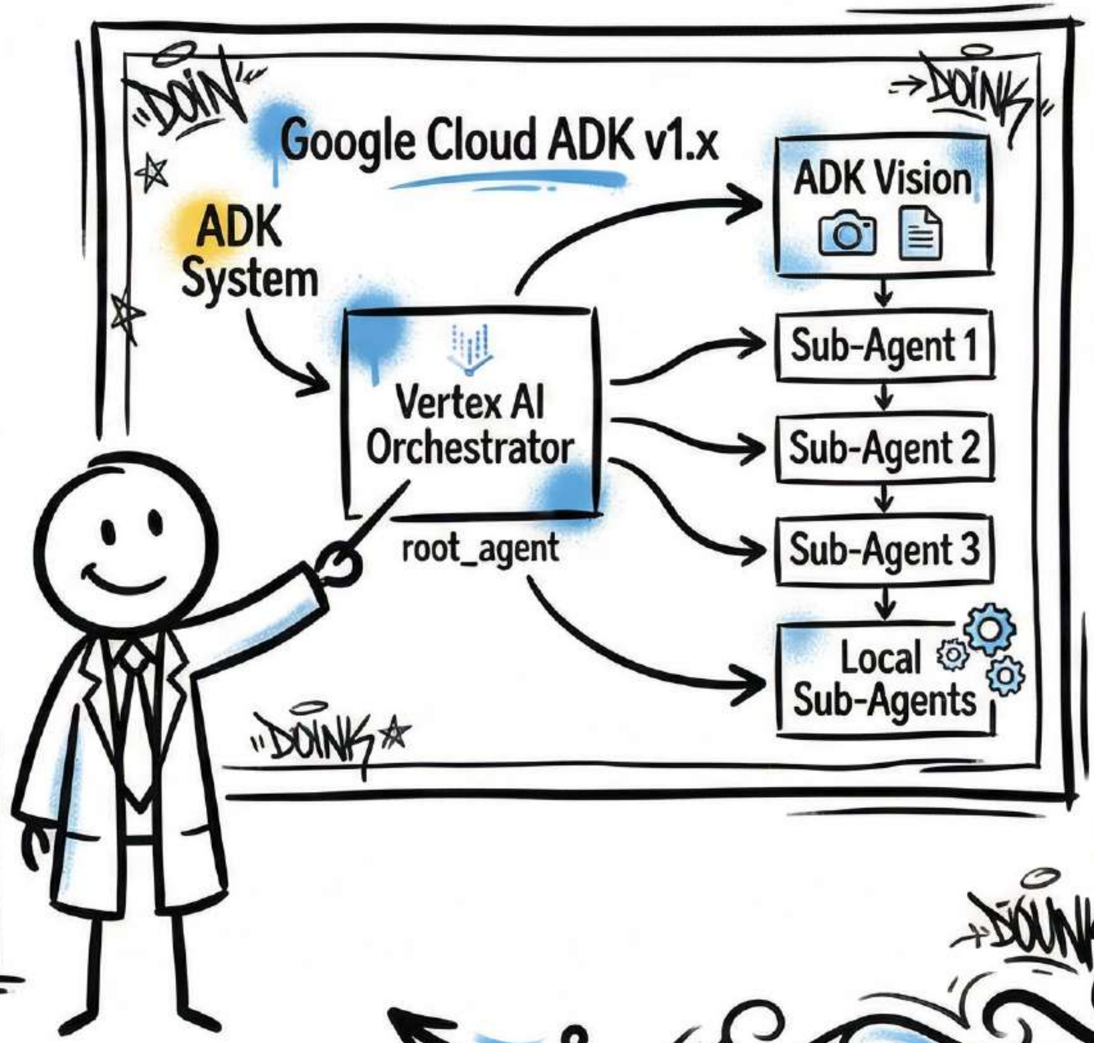
Brief about the idea

- **Project Name:** Career AI: Unified Multi-Agent Personal & Career Platform
- **Problem Statement:** Students and professionals face extreme "Context Switching" fatigue between task managers, disconnected APIs, and career guidance tools. Single-agent LLMs lack domain-specific tools, leading to generic AI hallucinations instead of actionable life strategy.
- **The Solution:** Career AI is a production-ready, ultra-resilient Multi-Agent orchestrator using Google ADK. It houses 3 deeply-specialized sub-agents (Tasks, Schedule, Career) and utilizes an intelligent "Dynamic Fast-Router" with an automatic 5-Layer Engine Fallback mechanism guaranteeing 100% uptime with sub-second text latency.



Meeting the Build Criteria

- **ADK Implementation:** Built using Google Cloud ADK v1.x for robust agent orchestration, effectively addressing the 'What You Must Build' criteria.
- **Specialized Routing:** A central Vertex AI Orchestrator (root_agent) analyzes user intent and routes payloads to specialized local sub-agents.
- **Multi-modal Capability:** Multi-modal queries (PDF/Images) natively bypass the router directly to ADK Vision layers for instant processing.
- **Future-Proofing:** Standardized for MCP (Model Context Protocol) to ensure boundless tool integration and future scalability.



Opportunities

- **Differentiation:** Unlike generic LLMs that hallucinate, Career AI uses domain-specific tools for actionable life strategies.
- **Problem Solving:** Intelligent dynamic routing between Fast Engines (Groq) and Deep Reasoning Engines (Gemini ADK) eliminates typical multi-agent lag.
- **USP:** The 'Dynamic Fast-Router' with a 5-Layer Engine Fallback mechanism guarantees 100% uptime and sub-second latency.

List of features offered by the solution

Autonomous Execution (Task Matrix)

Delegated tasks are handled independently from start to finish.

Career Trajectory (Career Nexus)

Predictive analytics for optimizing your professional path.

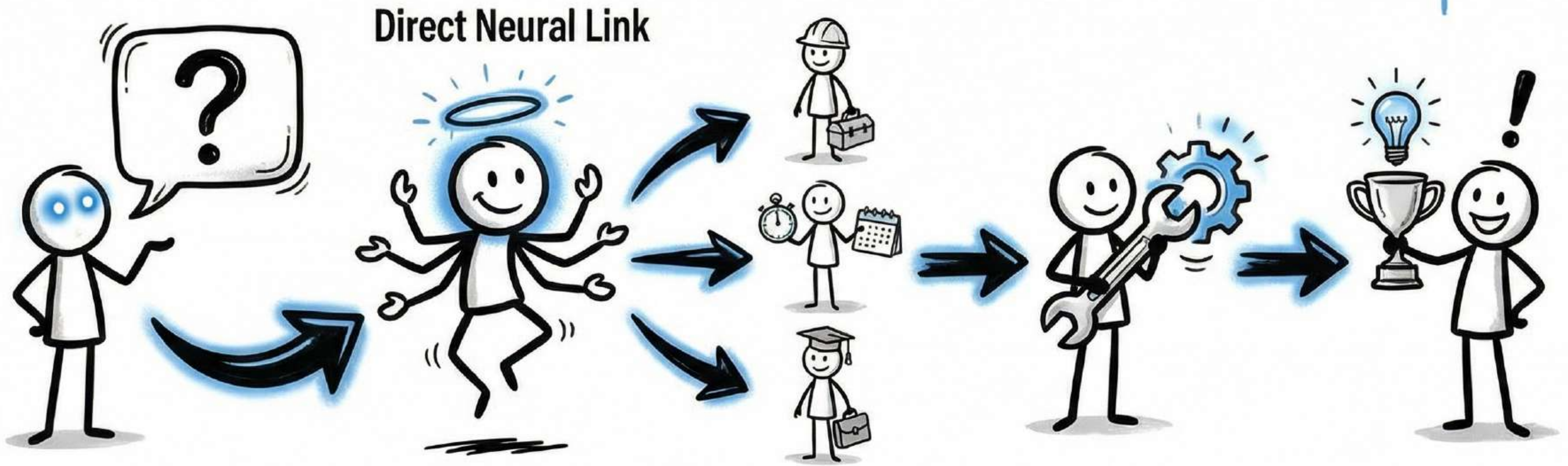
Context Retention (Deep Knowledge)

Long-term memory across sessions using advanced persistence.

Shielded Privacy (Uptime Guard)

Total isolation of local data with high-grade encryption and 100% resilience through Vertex AI and Llama 3.3 fallback engines.

Process Flow: How the Orchestrator Thinks



01
Input Synthesis

01
Orchestrator
Your request is parsed for intent and complex requirements.

02
Agent Selection
The orchestrator identifies the best specialists for the task (Tasks, Schedule, or Career).

03
Tool Calibration
Active MCP tools are connected and executed in parallel.

04
Final Resolution
The response is validated and delivered with full context.

Technologies to be used in the solution

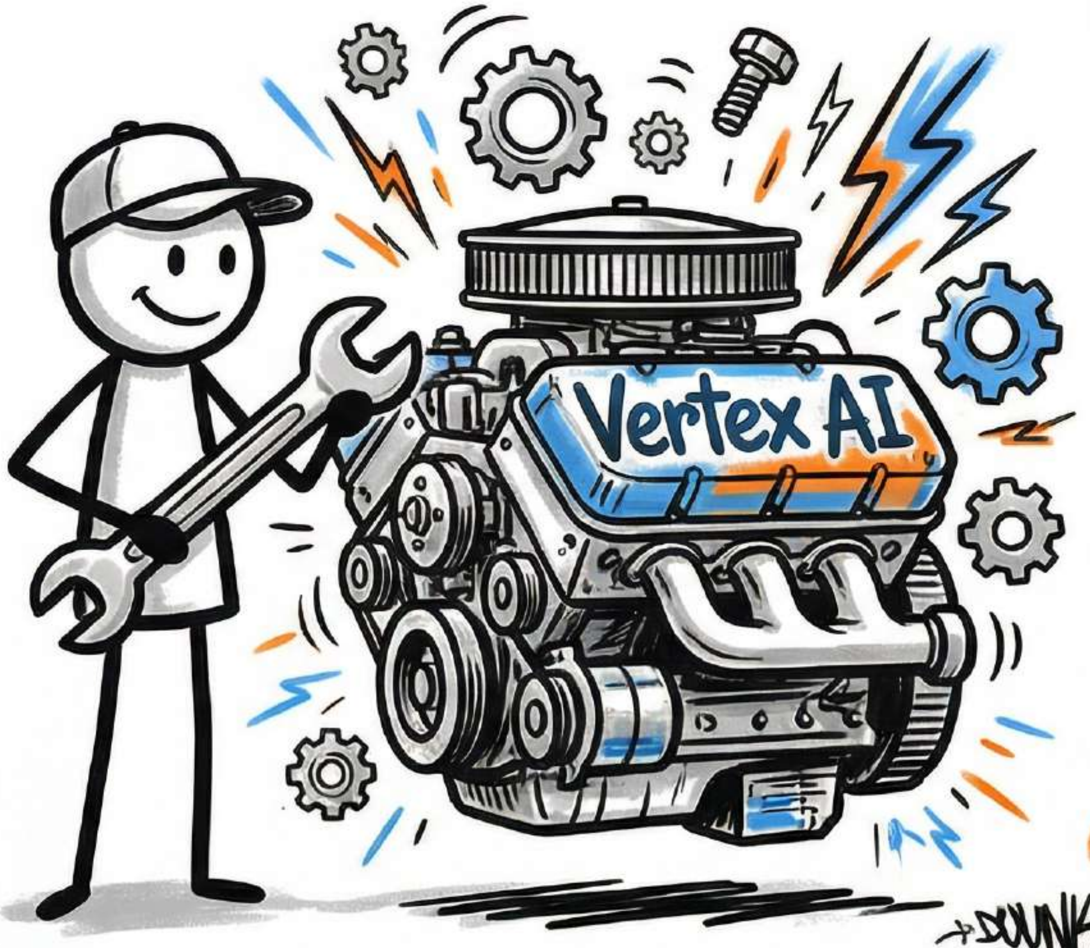
Frameworks:
- Google Cloud ADK v1.x, Starette Python Backend, Local SQLite.

Models:
- **Vertex AI** (Gemini 1.5 Pro / Flash) & Llama 3.3 Dynamic Fallback.

Infrastructure:
- Google Cloud Run (Dockerized Serverless Hosting), Cloud Build.

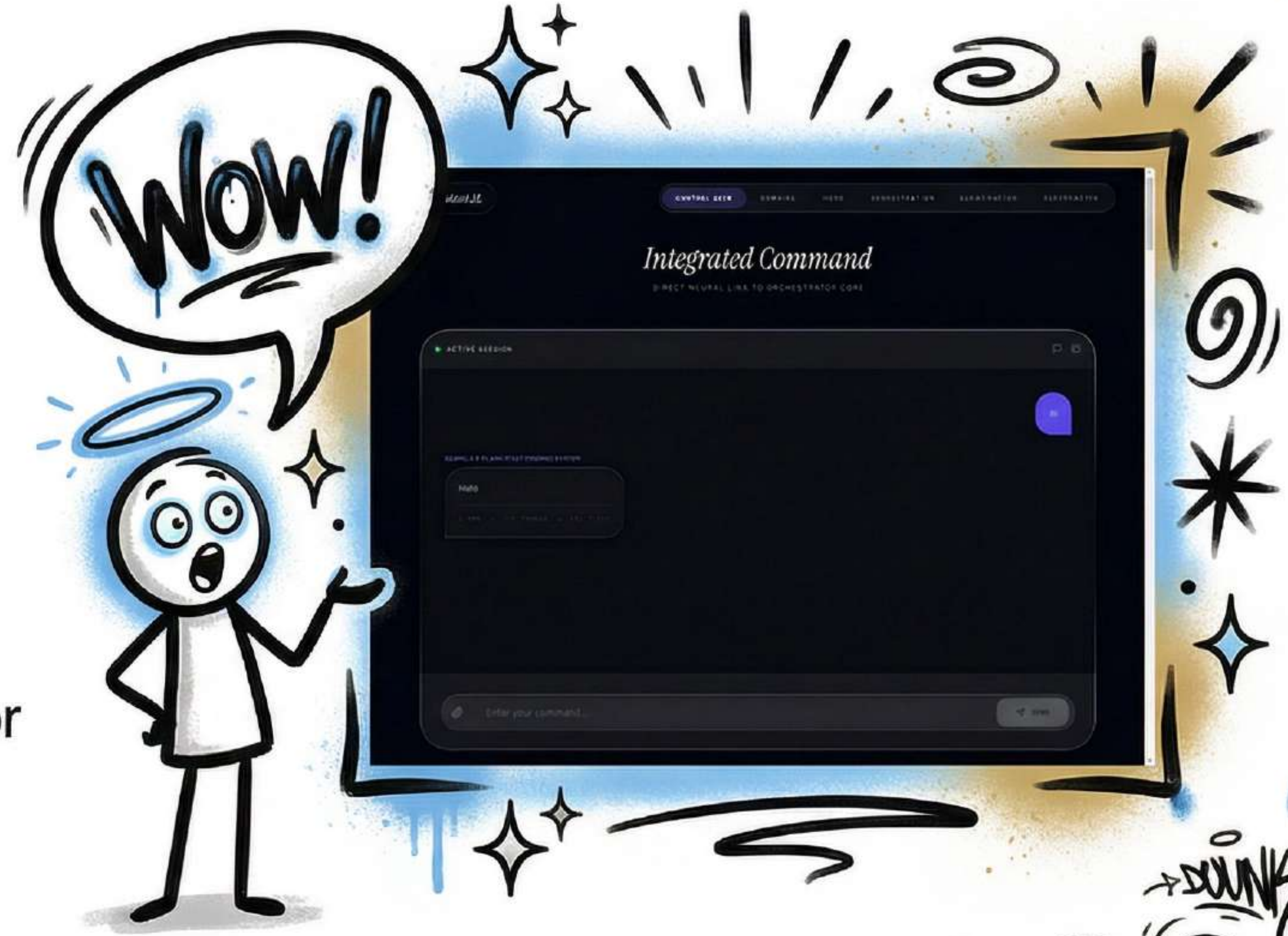
Latency Optimization: Dynamic routing between Fast Engines (Groq) and Deep Reasoning Engines (Gemini).

Tools Interface: Standardized for MCP (Model Context Protocol).



Snapshots of the prototype

- **Glassmorphism Interface**: A beautiful, futuristic UI designed for total task autonomy and professional evolution.
- **Integrated Command:** Direct neural link to the orchestrator core for seamless multi-agent interaction.
- **Responsive Design:** Optimized for high-performance across all devices with sub-second latency.



Learnings, Impact, & Link

Key Learnings:

- Intelligent dynamic routing between Fast Engines (Groq) and Deep Reasoning Engines (Gemini ADK) optimizes API token limits and server latency, eliminating multi-agent lag.

Real World Impact:

- A fully resilient, zero-downtime, personal life-and-career management hub that democratizes access to “always-on” hyper-personalized guidance.

Live Demo URLs:

- https://jtsvize-career-ai.hf.space/
- https://assistant-ai-uqu4.onrender.com/

Thank You!

- **Here is the GitHub link to the project:** https://github.com/Jtss-ux/assistant-ai/tree/master
- modified to run on Google Cloud, Render, Hugging Face, etc

Feel free to reach out for any inquiries or collaborations!